

A Geometric Correspondence for the Proton Charge Radius

From a Relational Projection of a Single Event

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Abstract

We present a relational geometric framework based on a postulate of uniqueness and a rule of minimal binary differentiation. Under these assumptions, one obtains, as an internal theorem, a discrete sequence

$$r_V = 2^{-\varepsilon}, \quad \varepsilon \in \mathbb{N}_{\geq 0},$$

and an associated arithmetic structure

$$M_\varepsilon = 2^\varepsilon - 1.$$

The first compositional closure occurs at $\varepsilon = 4$. If that structural position is assigned to the proton, the framework produces the projected scale

$$R_V = 4 \frac{\hbar}{m_p c} \simeq 0.841236 \text{ fm.}$$

We propose, as an electromagnetic bridge hypothesis, that this scale coincides with the proton root-mean-square charge radius. The equality $R_V = r_p^{\text{charge}}$ is not a theorem of the framework: it is the physical hypothesis that connects the relational geometry with the electromagnetic observable, and it is not yet derived from a microscopic theory of the current. Under this hypothesis, the scale R_V agrees with CODATA 2018 at the 0.02% level and remains within 1σ of the CODATA/NIST 2022 recommended value. Under an additional maximum-entropy ansatz for the projected density, one obtains

$$G_E(Q^2) = \exp\left(-\frac{Q^2 R_V^2}{6}\right),$$

with an independent prediction for the curvature,

$$C_4 = \frac{R_V^4}{72} \simeq 6.96 \times 10^{-3} \text{ fm}^4 \simeq 4.59 \text{ GeV}^{-4}.$$

The framework is presented as a falsifiable geometric correspondence, not as a replacement for quantum chromodynamics. The additional correspondences for the electron, the free neutron, baryonic trivalence, relational double covering, and mesons are formulated as limited-scope structural extensions.

1 Epistemic Status of the Framework

Before presenting the formal development, we distinguish four categories:

- **Postulate:** a foundational assumption or selection rule not derived within the framework.
- **Internal theorem:** a mathematical consequence of the postulates.
- **Bridge hypothesis:** an external physical identification connecting an internal quantity of the framework with an observable.
- **Falsifiable prediction:** a testable consequence of a theorem or of an additional ansatz.

Component	Status
Uniqueness of the event $\aleph$	Foundational postulate
Coherence of the present	Foundational postulate
Operationality of mathematics	Foundational postulate
Universal frequency $\omega_{\aleph}$	Foundational postulate
Image frequency $\omega_I = \omega_{\aleph} t_I$	Operational definition
Pythagorean identity $\omega_I^2 = \omega_V^2 + \omega_m^2$	Structural postulate
Minimal binary differentiation	Selection postulate
Quantization $r_V = 2^{-\varepsilon}$	Internal theorem, given the binary postulate
Mersenne structure $M_\varepsilon = 2^\varepsilon - 1$	Internal algebraic theorem
First compositional closure $\varepsilon^* = 4$	Mathematical theorem
Assignment $\varepsilon = 4 \rightarrow p$	Selection criterion, not a complete physical derivation
Relation $R_V = 4\bar{\lambda}_p$	Internal theorem if $\varepsilon = 4 \rightarrow p$ is accepted
Identification $R_V = r_p^{\text{charge}}$	Electromagnetic bridge hypothesis
Gaussian form of $G_E(Q^2)$	Low- Q^2 effective model under maximum entropy
Curvature $C_4 = R_V^4/72$	Falsifiable prediction of the Gaussian ansatz
Baryonic trivalence	Modest structural result
Double covering $\Phi = e^{i\phi/2}$	Conditional theorem $SU(2) \rightarrow SO(3)$
Exact relational non-duplication	Tautology under uniqueness; partial analogy with Pauli
Absolute derivation of binary differentiation	Open gap
Derivation of the bridge hypothesis	Main open gap
Heavy quarks	Open gap
Complete theory of mesons	Open gap

2 Introduction

The proton charge radius is a fundamental parameter of hadronic physics. Measurements using muonic hydrogen spectroscopy [1, 2], electron–proton scattering [3, 4], and electronic hydrogen spectroscopy [9] have converged, after the discrepancy known as the *proton-radius puzzle*, toward values close to 0.84 fm. The recommended values are $r_p = 0.8414(19)$ fm according to CODATA 2018 [5] and $r_p = 0.84075(64)$ fm according to CODATA/NIST 2022 [6].

In this work we propose a concrete geometric correspondence. The logical chain is

$$\text{uniqueness} + \text{binary differentiation} \implies r_V = 2^{-\varepsilon} \implies M_\varepsilon = 2^\varepsilon - 1.$$

The first compositional closure occurs at $\varepsilon = 4$. Assigning this structural position to the proton gives, internally,

$$R_V = 4 \frac{\hbar}{m_p c}.$$

The identification of this scale with the charge radius,

$$R_V = r_p^{\text{charge}},$$

is not a complete derivation from the geometric postulates. It is an electromagnetic bridge hypothesis that must be evaluated through additional consequences, in particular through the low- Q^2 curvature of the electric form factor $G_E(Q^2)$.

The framework is not intended to replace the Standard Model or quantum chromodynamics. Its purpose is more limited: to propose a relational architecture capable of organizing certain empirical regularities and producing a non-trivial numerical scale associated with the proton.

3 Foundational Postulates

Postulate 1 (Uniqueness of the origin). *The origin begins from a single event $\aleph$, characterized by a universal angular frequency $\omega_\aleph$ and a projection relation c . The totality of what is observable consists of images projected from $\aleph$ onto a relational space. The images are not ontologically separate entities: they are harmonics of $\aleph$, distinguishable by phase, harmonic depth, and relational context.*

Postulate 2 (Coherence of the present). *All projected images share a single present, synchronized with the current phase of the origin $\aleph$. The existence of an image consists in its coherence with that shared present. Loss of coherence corresponds to dissipation or to loss of stability as a free image.*

Postulate 3 (Operationality of mathematics). *The mathematical description of the model is operational: the equations define verifiable relations among measurable or potentially measurable quantities. The question of whether such equations describe, represent, or constitute the underlying ontological mechanism lies outside the formal scope of this work.*

4 Definitions

Definition 1 (Projected image). *A projected image of the origin $\aleph$ is a solution of the framework characterized by a pair*

$$I = (\varepsilon, \phi),$$

where $\varepsilon \in \mathbb{N}_{\geq 0}$ is the harmonic depth and $\phi \in [0, 2\pi)$ is the relational phase.

Definition 2 (Harmonic image frequency). *For each image I , its characteristic harmonic frequency is*

$$\omega_I \equiv \omega_\aleph t_I,$$

where t_I is the coupling factor of the image with the shared present. For images with observable mass m_I , we use the operational Compton relation

$$\omega_I = \frac{m_I c^2}{\hbar}.$$

In this work, t_I is fixed by the observable mass; an independent derivation of t_I from the structure of the framework remains open.

Definition 3 (Modal decomposition). *The harmonic image frequency decomposes into two orthogonal components:*

$$\omega_I^2 = \omega_V^2 + \omega_m^2,$$

where ω_V is the projected mode and ω_m is the confined mode. The two components are not separate fundamental frequencies: they are orthogonal components of a single characteristic frequency.

The component ω_V represents what can be projected onto external references and generate effective observables, such as size scales or couplings. The component ω_m represents what sustains internal coherence and is not projected directly.

Definition 4 (Modal weights). *The modal decomposition is parametrized by weights*

$$r_V = \frac{W_V}{W_V + W_m}, \quad r_m = \frac{W_m}{W_V + W_m},$$

with

$$r_V + r_m = 1,$$

and

$$\omega_V = \omega_I \sqrt{r_V}, \quad \omega_m = \omega_I \sqrt{r_m}.$$

Definition 5 (Relational radii). *To each modal component we associate a characteristic scale:*

$$R_V = \frac{c}{\omega_V}, \quad R_m = \frac{c}{\omega_m}.$$

5 Binary Differentiation and Quantization

Postulate 4 (Minimal binary differentiation). *Every elementary relational distinction between an image and the origin separates information into two orthogonal and exhaustive channels. Each distinction preserves the quadratic sum of the components and halves the projected fraction.*

This postulate formalizes the idea that the individuation of an image proceeds through successive elementary distinctions. The strong assumption is not the subsequent arithmetic, but rather that the minimal admissible distinction is binary and that its effect on the projected fraction is a halving rule.

Theorem 1 (Binary quantization). *If an image requires ε independent relational distinctions to close its identity with respect to the origin, then*

$$r_V = 2^{-\varepsilon}, \quad \omega_V = \omega_I 2^{-\varepsilon/2}.$$

Proof. Let $r_V(k)$ be the projected fraction after k distinctions, with $r_V(0) = 1$. By the postulate of minimal binary differentiation,

$$r_V(k+1) = \frac{r_V(k)}{2}.$$

By induction,

$$r_V(\varepsilon) = 2^{-\varepsilon}.$$

The relation for ω_V follows from $\omega_V = \omega_I \sqrt{r_V}$. □

6 Mersenne Structure and Proton Selection

Theorem 2 (Mersenne structure). *For a coherent image with depth ε ,*

$$\frac{\omega_m^2}{\omega_V^2} = 2^\varepsilon - 1 \equiv M_\varepsilon.$$

Proof. From the modal decomposition,

$$\omega_m^2 = \omega_I^2 - \omega_V^2.$$

Using $\omega_V^2 = \omega_I^2 2^{-\varepsilon}$, one obtains

$$\frac{\omega_m^2}{\omega_V^2} = \frac{\omega_I^2(1 - 2^{-\varepsilon})}{\omega_I^2 2^{-\varepsilon}} = 2^\varepsilon - 1.$$

□

6.1 Low-order spectrum

ε	M_ε	Factorization	New primes	Tentative correspondence
0	0	—	—	null mode
1	1	unit	—	symmetric case
2	3	prime	{3}	electron
3	7	prime	{7}	free neutron
4	15	$3 \cdot 5$	{5}	proton
5	31	prime	{31}	—
6	63	$3^2 \cdot 7$	—	—
7	127	prime	{127}	—

6.2 First compositional closure

Theorem 3 (First compositional closure). *Let ε^* be the smallest integer for which M_{ε^*} is composite and its factorization contains at least one prime previously present in the sequence $\{M_\varepsilon\}_{\varepsilon < \varepsilon^*}$ and at least one new prime. Then*

$$\varepsilon^* = 4.$$

Proof. For $\varepsilon = 0, 1$ the values are trivial. For $\varepsilon = 2$ and $\varepsilon = 3$, one obtains primes, 3 and 7. The first composite value with inherited and new prime content is

$$M_4 = 15 = 3 \cdot 5.$$

□

The assignment $\varepsilon = 4 \rightarrow p$ selects a structural position, not a complete physical identity. The framework does not yet derive proton charge, baryon number, color confinement, or the complete *uud* structure. The assignment is proposed because $\varepsilon = 4$ is the first compositional closure and because, under the bridge hypothesis below, it yields the observed electromagnetic scale of the proton. Its validity therefore depends on independent tests, especially the curvature of the electric form factor.

7 Projected Scale of the Proton

Theorem 4 (Relational scale of the proton). *For the image $\varepsilon = 4$ assigned to the proton,*

$$R_V = 4\bar{\lambda}_p = 4\frac{\hbar}{m_p c} \simeq 0.841236 \text{ fm.}$$

Proof. For $\varepsilon = 4$,

$$\omega_V = \omega_I 2^{-4/2} = \frac{\omega_I}{4}.$$

For the proton,

$$\omega_I = \omega_p = \frac{m_p c^2}{\hbar}.$$

Therefore,

$$\omega_V = \frac{m_p c^2}{4\hbar}.$$

Using $R_V = c/\omega_V$,

$$R_V = \frac{c}{m_p c^2/(4\hbar)} = 4\frac{\hbar}{m_p c} = 4\bar{\lambda}_p.$$

□

The result above is a reduced Compton scale multiplied by 4. By itself it is not yet the proton charge radius. Equality with the charge radius requires the bridge hypothesis of the following section.

8 Electromagnetic Bridge Hypothesis

The experimental observable r_p^{charge} is defined through the slope of the proton electric form factor at the origin:

$$\langle r^2 \rangle_E = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0}.$$

This quantity belongs to the proton's effective electromagnetic current, not directly to the internal geometry defined so far.

Hypothesis 1 (Electromagnetic bridge). *Let J_{eff}^μ be the effective electromagnetic current associated with the proton image. In the low-momentum-transfer effective limit, the projected component V determines the second moment of the electric distribution:*

$$\langle r^2 \rangle_E = R_V^2.$$

Equivalently,

$$r_p^{\text{charge}} = R_V.$$

The equality $R_V = r_p^{\text{charge}}$ is not a theorem of the framework. It is the physical hypothesis that connects the relational scale with the electromagnetic observable. Without this hypothesis, the framework produces a geometric scale close to the proton radius but does not establish its physical identity.

The central open question is: *why should the V mode couple to the photon as electric charge while the m mode does not?* A complete resolution would require constructing J_{eff}^μ from the relational structure, proving that its form factor satisfies $\langle r^2 \rangle_E = R_V^2$, and identifying V as the geometric carrier of the electromagnetic coupling.

8.1 Conditional numerical comparison

Under the bridge hypothesis,

$$r_p^{\text{model}} = 4 \frac{\hbar}{m_p c} \simeq 0.841236 \text{ fm.}$$

Compared with CODATA 2018,

$$r_p^{\text{exp}} = 0.8414(19) \text{ fm,}$$

the relative difference is approximately

$$\frac{r_p^{\text{model}} - r_p^{\text{exp}}}{r_p^{\text{exp}}} \simeq -1.9 \times 10^{-4}.$$

Compared with CODATA/NIST 2022,

$$r_p^{\text{exp}} = 0.84075(64) \text{ fm,}$$

the absolute deviation is

$$|0.841236 - 0.84075| \text{ fm} \simeq 0.000486 \text{ fm,}$$

equivalent to approximately 0.76σ .

The associated dimensionless constant is

$$\kappa \equiv \frac{m_p c r_p}{\hbar}.$$

The framework, conditioned on the bridge hypothesis, gives

$$\kappa_{\text{model}} = 4.$$

Experimentally one obtains approximately $\kappa_{\text{exp}} = 4.0008$ using CODATA 2018 and $\kappa_{\text{exp}} = 3.9977$ using CODATA/NIST 2022.

9 Projected Density and Form Factor

9.1 Maximum-entropy ansatz

The bridge hypothesis fixes only the second moment. It does not determine the full functional form of the projected relational density $\rho_V(\vec{r})$. To obtain a minimal functional form, we introduce a maximum-entropy ansatz.

Theorem 5 (Minimal relational density). *Let $\rho_V(\vec{r})$ be an effective density associated with the projected mode, subject to*

$$\begin{aligned} \int \rho_V(\vec{r}) d^3r &= 1, \\ \int r^2 \rho_V(\vec{r}) d^3r &= R_V^2, \\ \rho_V(\vec{r}) &\geq 0. \end{aligned}$$

The maximum-entropy distribution under these constraints is

$$\rho_V(\vec{r}) = \left(\frac{3}{2\pi R_V^2} \right)^{3/2} \exp\left(-\frac{3r^2}{2R_V^2} \right).$$

Proof. Functional maximization of

$$S[\rho] = - \int \rho \ln \rho d^3r$$

with Lagrange multipliers for normalization and the second moment produces a Gaussian. Isotropy follows from the fact that the only non-trivial geometric constraint is radial. $\square$

This result does not independently prove that the physical second moment of charge is R_V^2 ; it presupposes it through the bridge hypothesis. The Gaussian form is the least informative distribution compatible with that constraint.

9.2 Effective form factor

In the low- Q^2 effective regime, using the non-relativistic Fourier transform as an approximation,

$$G_E(Q^2) = \exp\left(-\frac{Q^2 R_V^2}{6}\right).$$

Its expansion is

$$G_E(Q^2) = 1 - \frac{Q^2 R_V^2}{6} + \frac{Q^4 R_V^4}{72} + O(Q^6).$$

Comparing with

$$G_E(Q^2) = 1 - \frac{Q^2 \langle r^2 \rangle_E}{6} + C_4 Q^4 + \dots,$$

one obtains

$$C_4 = \frac{R_V^4}{72}.$$

For $R_V = 0.841236$ fm,

$$C_4 \simeq 6.96 \times 10^{-3} \text{ fm}^4.$$

If Q^2 is expressed in GeV^2 ,

$$C_4^{[\text{GeV}^{-4}]} = C_4^{[\text{fm}^4]} (\hbar c)^{-4} \simeq 4.59 \text{ GeV}^{-4},$$

using $\hbar c \simeq 0.1973 \text{ GeV fm}$.

This curvature is the independent falsifiable prediction of the Gaussian ansatz. If low- Q^2 electron-proton scattering data require a substantially different curvature, the maximum-entropy ansatz is refuted, although the scale $R_V = 4\lambda_p$ could survive either as a coincidence or as part of a non-Gaussian density.

Preliminary empirical anchoring. Extractions of the curvature of $G_E(Q^2)$ are not yet universal, because they depend on the parametrization used, the Q^2 range, and the treatment of experimental normalizations. Even so, published values for the effective quadratic coefficient lie on the scale of several 10^{-3} to slightly above 10^{-2} fm^4 . For example, Hill and Paz report reference values of $0.011(4) \text{ fm}^4$ and $0.014(4) \text{ fm}^4$ for the Taylor quadratic coefficient used in radius extractions [7]. A recent integral analysis of electric form-factor data obtained $\langle r^4 \rangle = 1.47(25)_{\text{stat}}(45)_{\text{sys}} \text{ fm}^4$, which in the non-relativistic expansion

$$G_E(Q^2) = 1 - \frac{Q^2 \langle r^2 \rangle}{6} + \frac{Q^4 \langle r^4 \rangle}{120} + \dots$$

corresponds to $C_4 \simeq 1.23 \times 10^{-2} \text{ fm}^4$ [14]. The value predicted here,

$$C_4^{\text{model}} = 6.96 \times 10^{-3} \text{ fm}^4,$$

falls within that empirical scale. This does not constitute a confirmation of the Gaussian ansatz, but it does indicate that the ansatz is not ruled out by the order of magnitude of existing extractions. The known tension among different low- Q^2 extractions of $G_E(Q^2)$ reinforces the need for a dedicated curvature test with non-Gaussian parametrizations and explicit control of systematic effects [12, 13].

10 Additional Correspondences: Limited Scope

The correspondences in this section do not automatically use the proton electromagnetic bridge hypothesis. They should therefore not be read as predictions of charge radii. They are internal coherence scales associated with tentative structural assignments.

10.1 Electron

For $\varepsilon = 2$, tentatively assigned to the electron,

$$R_V^{(e)} = 2\bar{\lambda}_e, \quad R_m^{(e)} = \frac{2}{\sqrt{3}}\bar{\lambda}_e.$$

Here $R_V^{(e)}$ does not represent a charge radius or a classical spatial extension of the electron. It is a coherence/projection scale related to the Compton wavelength, where the notion of a classical point particle ceases to be adequate and pair creation becomes kinematically relevant. The interpretive asymmetry between proton and electron is an explicit limitation of the present framework.

10.2 Free neutron

For $\varepsilon = 3$,

$$R_V^{(n)} = 2\sqrt{2}\bar{\lambda}_n, \quad R_m^{(n)} = \frac{2\sqrt{2}}{\sqrt{7}}\bar{\lambda}_n.$$

The harmonic signature $M_3 = 7$ is prime. This arithmetic rigidity is interpreted as a structural analogy with the instability of the free neutron, but not as a causal mechanism. Neutron beta decay depends on the weak interaction and on the mass difference among neutron, proton, electron, and antineutrino.

11 Baryonic Trivalence

Definition 6 (Confined subharmonics). *For an image with composite M_ε , we define subharmonics of the confined mode by*

$$\omega_m^{(d)} = \frac{\omega_m}{d}, \quad d \mid M_\varepsilon.$$

For $\varepsilon = 4$,

$$M_4 = 15 = 3 \cdot 5.$$

The first non-trivial divisor is $d = 3$. We propose that the subharmonic $\omega_m/3$ defines a substructure of three correlated modes with phases separated by $2\pi/3$. The associated equivalent mass is

$$m_q = \frac{\sqrt{15}}{12}m_p \simeq 302.83 \text{ MeV}/c^2.$$

This value lies on the phenomenological scale of light u/d constituent masses, typically of order $300 \text{ MeV}/c^2$. The correspondence is modest: it does not derive color, QCD confinement, fractional charge, or gluon dynamics.

12 Double Covering and Spinorial Analogy

12.1 Internal orientation

For images with $\varepsilon \geq 2$, we introduce a discrete orientation

$$\sigma = \text{sgn}(\sin \phi) \in \{+1, -1\}.$$

For $\varepsilon \in \{0, 1\}$, the hemispheres are geometrically equivalent and no non-trivial internal orientation is introduced.

Theorem 6 (Relational double covering). *Let an image with $\varepsilon \geq 2$ preserve an internal orientation σ that is not directly observable, and whose inversion changes the relational state. If the observable space of orientations is $SO(3)$, then the minimal space covering the internal orientations is $SU(2)$, and the relational phase transforms as*

$$\Phi = e^{i\phi/2}.$$

Under a frame rotation by θ ,

$$\Phi \mapsto e^{i\theta/2}\Phi.$$

Proof. The universal double cover of $SO(3)$ is $SU(2)$, with homomorphism

$$\pi : SU(2) \rightarrow SO(3)$$

and kernel $\{+I, -I\}$. A 2π rotation in $SO(3)$ corresponds to multiplication by $-I$ in $SU(2)$. Therefore, a phase living in the double cover acquires the behavior $e^{i\theta/2}$. $\square$

This result is not equivalent to a complete derivation of physical spin $1/2$. Such a derivation requires representations of the Poincaré group, relativistic Hilbert spaces, quantum fields, and the spin–statistics theorem. The present framework only reproduces an analogous minimal geometric condition: double covering.

12.2 Boson–fermion analogy

Images with $\varepsilon \in \{0, 1\}$ do not require a non-trivial double cover; their phase may be taken as $\Phi = e^{i\phi}$ and returns to itself under 2π . Images with $\varepsilon \geq 2$ admit the double-cover structure described above. The framework therefore produces a relational distinction analogous to the boson–fermion separation, but not a complete derivation of quantum statistics.

12.3 Relational non-duplication

Two images with the same depth $\varepsilon \geq 2$ cannot simultaneously share the same phase ϕ , the same internal orientation σ , and the same local relational context. If they coincided in all relational identifiers, they would be a single projection, not two.

This is an identity property within the framework, not the full Pauli exclusion principle. It may be understood as a geometric condition analogous to the non-duplication of identical states, but Pauli requires antisymmetry of fermionic states in Hilbert space and second quantization.

13 Partial Mesonic Criterion

Mesons are not treated here as complete solutions of the scheme $r_V = 2^{-\varepsilon}$. This section formulates a programmatic criterion, not a theory of mesons.

Definition 7 (Calibrated mesonic state). *A mesonic state with mass m_M has total frequency*

$$\omega_I^{(M)} = \frac{m_M c^2}{\hbar}.$$

Its projected component is calibrated with the protonic component

$$\omega_V^{(p)} = \frac{\omega_p}{4}.$$

The confined mesonic component is defined by

$$\left(\omega_I^{(M)}\right)^2 = \left(\omega_V^{(p)}\right)^2 + \left(\omega_m^{(M)}\right)^2.$$

If

$$\omega_I^{(M)} < \omega_V^{(p)},$$

then $\omega_m^{(M)}$ is imaginary. Mathematically this converts oscillatory behavior into exponential behavior. Relationally, it is interpreted either as loss of free coherence or as existence in a complementary, non-projected modality.

Meson	m_M (MeV/ c^2)	$\omega_I^{(M)}$ (rad/s)	Category
$\pi^\pm$	139.57	2.12×10^{23}	sub-baryonic
π^0	134.98	2.05×10^{23}	sub-baryonic
$K^\pm$	493.68	7.50×10^{23}	supra-baryonic
η	547.86	8.32×10^{23}	supra-baryonic
ρ	775.26	1.18×10^{24}	supra-baryonic

The choice of $\omega_V^{(p)}$ as reference is a calibration, not a derivation. Meson masses are used as inputs and lifetimes are not predicted. A complete theory should formulate mesons as relational dipoles between an image and an anti-image, with an internal cancellation rule that is not yet developed.

14 Remaining Gaps and Falsification Program

14.1 Absolute derivation of binary differentiation

The postulate of binary differentiation turns the quantization $r_V = 2^{-\varepsilon}$ into an internal theorem, but the postulate itself remains underived. One possible route would be to obtain it from a topological condition of self-image or phase closure of the origin cycle $\aleph$.

14.2 Derivation of the electromagnetic bridge hypothesis

This is the main gap. The framework must construct an effective current

$$J_{\text{eff}}^\mu$$

such that

$$\langle r^2 \rangle_E = -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0} = R_V^2,$$

and it must explain why the V mode couples to the photon as electric charge while the m mode does not.

14.3 Heavy quarks

The trivalent substructure produces a light constituent mass, but it does not generate the s, c, b, t generations or the complete baryon spectrum. A viable extension would require a quadratic composition rule for effective weights with minimal calibration and independent predictions.

14.4 Mesons

The current mesonic criterion classifies states from a protonic calibration. A complete theory must derive masses, lifetimes, decay channels, and coupling structure.

14.5 Falsification protocol

The most direct test of the framework, in its present form, is not merely the radius coincidence, but the curvature:

$$C_4 = \frac{R_V^4}{72}.$$

The suggested experimental procedure is to compare this curvature with low- Q^2 extractions of the proton electric form factor, using parametrizations that do not impose a Gaussian form in advance. As a scale guide, existing analyses place effective curvature coefficients at order 10^{-3} – 10^{-2} fm^4 , with still significant uncertainties and model dependence [7, 14, 12]. If a controlled low- Q^2 extraction is incompatible with $C_4 = 6.96 \times 10^{-3} \text{ fm}^4$, or equivalently with 4.59 GeV^{-4} , within reasonable uncertainties, then the maximum-entropy ansatz is falsified.

15 Conclusions

The framework presented here internally derives a discrete sequence

$$r_V = 2^{-\varepsilon}$$

and an associated structure

$$M_\varepsilon = 2^\varepsilon - 1.$$

The first compositional closure occurs at $\varepsilon = 4$. If that structural position is assigned to the proton, one obtains as an internal theorem

$$R_V = 4\bar{\lambda}_p = 4\frac{\hbar}{m_p c} \simeq 0.841236 \text{ fm}.$$

The central physical claim has two epistemologically distinct steps:

1. **Internal theorem:** given the framework and the assignment $\varepsilon = 4 \rightarrow p$, one obtains $R_V = 4\bar{\lambda}_p$.
2. **Bridge hypothesis:** the scale R_V coincides with the proton charge radius, $R_V = r_p^{\text{charge}}$.

The equality $R_V = r_p^{\text{charge}}$ is not a theorem of the framework. It is the physical hypothesis that connects the relational geometry with the electromagnetic observable. Under that hypothesis, the scale obtained agrees with CODATA values within experimental uncertainty. Under the additional maximum-entropy ansatz, the model produces a falsifiable prediction for the curvature of the form factor:

$$C_4 = \frac{R_V^4}{72} \simeq 6.96 \times 10^{-3} \text{ fm}^4 \simeq 4.59 \text{ GeV}^{-4}.$$

The additional correspondences with the electron, neutron, baryonic trivalence, relational double covering, and mesons should be understood as limited-scope structural extensions, not as a complete reconstruction of the Standard Model. The most robust result of the work is not a final derivation of the proton radius, but a quantitative and falsifiable geometric correspondence that identifies precisely where the theory must be closed: in the derivation of the effective electromagnetic current.

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References

- [1] R. Pohl et al., “The size of the proton”, *Nature* **466**, 213–216 (2010).
- [2] A. Antognini et al., “Proton structure from the measurement of 2S–2P transition frequencies of muonic hydrogen”, *Science* **339**, 417–420 (2013).
- [3] J. C. Bernauer et al., “High-precision determination of the electric and magnetic form factors of the proton”, *Phys. Rev. Lett.* **105**, 242001 (2010).
- [4] W. Xiong et al., “A small proton charge radius from an electron–proton scattering experiment”, *Nature* **575**, 147–150 (2019).
- [5] E. Tiesinga, P. J. Mohr, D. B. Newell and B. N. Taylor, “CODATA recommended values of the fundamental physical constants: 2018”, *Rev. Mod. Phys.* **93**, 025010 (2021).
- [6] NIST/CODATA recommended values 2022, “Proton rms charge radius”, <https://physics.nist.gov/cgi-bin/cuu/Value?rp>.
- [7] R. J. Hill and G. Paz, “Model-independent extraction of the proton charge radius from electron scattering”, *Phys. Rev. D* **82**, 113005 (2010).
- [8] G. A. Miller, “Charge densities of the neutron and proton”, *Phys. Rev. Lett.* **99**, 112001 (2007).
- [9] N. Bezginov et al., “A measurement of the atomic hydrogen Lamb shift and the proton charge radius”, *Science* **365**, 1007–1012 (2019).
- [10] D. Djukanovic et al., “Lattice QCD calculation of the nucleon electromagnetic form factors”, *Phys. Rev. D* **103**, 094522 (2021).
- [11] K. Abe et al. (Super-Kamiokande Collaboration), “Search for proton decay via $p \rightarrow e^+ \pi^0$ and $p \rightarrow \mu^+ \pi^0$ ”, *Phys. Rev. D* **95**, 012004 (2017).

- [12] J. Zhou, V. Khachatryan, H. Gao, S. Gorbaty and D. W. Higinbotham, “Understanding the systematic differences in extractions of the proton electric form factors at low Q^2 ”, *Phys. Rev. C* **106**, 065505 (2022).
- [13] W. Xiong and C. Peng, “Proton electric charge radius from lepton scattering”, *Universe* **9**, 182 (2023).
- [14] M. Atoui, M. B. Barbaro, M. Hoballah, C. Keyrouz, M. Lassaut, D. Marchand, G. Quéméner and E. Voutier, “Determination of the moments of the proton charge density”, arXiv:2304.13521v5 (2025).
- [15] E. T. Jaynes, “Information theory and statistical mechanics”, *Phys. Rev.* **106**, 620–630 (1957).